Spinal Motion Restriction

Mechanism of Injury
Suggests spinal Injury may be possible

Negative Mechanism

Positive Mechanism Or Unsure

- Apply manual Stabilization until exam complete
- Remove Helmet (if worn)

Spinal Pain or Tenderness? — YES

NO

Motor and Sensory Exam ABNORMAL? — YES

NO

Is the Patient Unreliable? — YES

NO

Spinal Motion Restriction **Not Required**

Spinal Motion Restriction **Required**

Unreliable Patient Has:
- Acute Stress Reaction
- Head/Brain Injury
- Altered Mental Status
- Intoxication with Drugs/Alcohol
- Other Distracting Injuries

Positive Mechanisms of Injury
- High-Speed MVC
- Falls >Three Times Patient's Height
- Axial load
- Diving Accidents
- Sports Injuries to Head or Neck
- Unconscious Trauma Patient

Patient Care Goals

1. Identify patients at risk of spinal injury and apply appropriate spinal motion restriction (SMR) to prevent further neurological damage.
2. Minimize discomfort and complications associated with SMR while ensuring patient safety during transport.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients with suspected spinal injury based on mechanism of injury, including:
    - Road Traffic Accidents
    - Falls
    - Axial load
    - Diving Accidents
    - Sports Injuries to Head or Neck
- Unconscious Trauma patients.

Exclusion Criteria

- Patients with no trauma history or clinical findings suggestive of spinal injury (e.g., isolated extremity injury without axial loading).
- Patients with penetrating trauma to the head, neck, or torso without neurological deficits, where SMR may delay critical interventions.

Key Patient Assessment Points

1. **Mechanism of Injury**
    - Identify high-risk mechanisms (e.g., axial loading, hyperflexion/hyperextension, or rotational forces) suggestive of spinal injury.
    - Confirm details of trauma (e.g., height of fall, ejection from vehicle, or impact velocity).
2. **Spinal Assessment**
    - Palpate the cervical, thoracic, and lumbar spine for midline tenderness, deformity, or step-offs.
    - Assess for pain with movement or spontaneous complaints of neck/back pain.
3. **Neurological Evaluation**
    - Check for neurological deficits (e.g., numbness, tingling, or weakness in extremities).
    - Perform motor and sensory exams (e.g., grip strength, foot dorsiflexion, light touch sensation).
    - Assess for priapism, loss of bowel/bladder control, or other signs of spinal cord injury.
4. **SMR Clearance Criteria (NEXUS Criteria)**
    - Evaluate for safe clearance of SMR using NEXUS Criteria:
        - No midline cervical tenderness.
        - No neurological deficits.
        - No altered mental status or intoxication.
        - No distracting injuries (e.g., severe pain elsewhere).
        - Low-risk mechanism of injury.
    - A positive result to any of the above criteria requires SMR.
5. **Determine Triage Category**
    - Determine Triage Category as per Patient Triage Categories CPG.
        - All patients who require Spinal Motion Restriction are yellow priority (as a minimum).

Patient Management & Interventions

1. **Primary Survey**
    - Ensure airway patency, adequate breathing, and circulation while maintaining manual cervical spine stabilization.
    - Control catastrophic external bleeding if present.
2. **Secondary Survey**
    - Conduct a focused spinal and neurological examination to confirm need for SMR.

- Obtain history of trauma mechanism and presenting symptoms (e.g., pain, numbness).
- Reassess vital signs and neurological status:

3. **Consider Red Flags**
   - Neurological deficits (e.g., paralysis, sensory loss, or priapism) indicating spinal cord injury.
   - Midline tenderness or deformity of the spine suggesting fracture or instability.
   - Hypotension with bradycardia (neurogenic shock) suggestive of high cervical injury.
4. **Spinal Motion Restriction**
   - **Spinal Motion Restriction Goals**
     - The goal of spinal motion restriction is to limit movement of the c-spine.
     - This is best achieved through the least invasive means to ensure that the patient does not move.
     - Use of extrication devices (such as a KED) may increase movement and should be used with caution.
     - The use of a c-collar may cause neck movement and should be used with caution to prevent unnecessary movement.
     - Where possible, the patient should be assisted onto the stretcher rather than using an extrication device.
     - Back boards (LBB) should be used as extrication devices only – patients should not be routinely transported to hospital on a back board.
     - Scoop stretchers are preferred to facilitate patient movement – where possible, the scoop stretcher should be removed once the patient is on the ambulance stretcher.
     - It is permitted to leave the LBB or scoop stretcher in place if immediate interventions (such as airway management) are required.
   - **Apply Manual In-Line Stabilization**
     - Maintain manual cervical spine stabilization until a cervical collar is applied.
     - Use jaw thrust for airway opening if cervical injury is suspected.
   - **Apply Cervical Collar**
     - Apply a rigid cervical collar for patients with suspected cervical spine injury or inability to clear SMR.
     - Ensure proper sizing to avoid discomfort or airway obstruction.
       - Contraindications to C-Collar Use:
         - Severe traumatic brain injuries (as it impedes cerebral venous damage, further increasing ICP).
         - Inability to maintain an effective airway because of c-collar placement
   - **Spinal Board or Scoop Stretcher**
     - Use a long spinal board or scoop stretcher to extricate the patient and/or move the patient on to the ambulance stretcher.
     - Log roll with multiple providers to maintain neutral alignment during transfer
   - **Padding and Comfort**
     - Pad spaces under the head and lumbar spine to maintain neutral alignment and reduce pressure sores.
5. **Walking Patients**
   - Consider application a c-collar (if indicated as per SMR assessment)
     - Routine application of a c-collar is not required.
   - Assist the patient to sit on stretcher with the head of the stretcher elevated.
   - Lay the patient supine by lowering the head of the stretcher.
   - Support the patients' head through using the head plate and head supports from the LBB.
     - Forehead and chinstraps are not required
   - Secure the patients torso using stretcher seat belts to avoid a fulcrum/pivot effect on the neck.
6. **Extrication from a Vehicle**
   - Apply a c-collar prior to extrication (if indicated as per SMR assessment)
   - Assist the patient **to self-extricate from the vehicle if they can do so:**
     - A LBB or scoop is not required.
     - Position the stretcher close to the vehicle.
     - Assist the patient to self-extricate and sit on stretcher with the head of the stretcher elevated.
     - Lay the patient supine by lowering the head of the stretcher.

        - Support the patients' head through using the head plate and head supports from the LBB.
            - Forehead and chinstraps are not required.
            - Secure the patients torso using stretcher seat belts to avoid a fulcrum/pivot effect on the neck.
    - **If the patient is unable to move:**
        - Use an extrication device. (Scoop/LBB etc. as appropriate).

7. **Agitated Patients**
    - If c-collars are causing the patient to become agitated and not remain still on the stretcher, it is appropriate to remove the c-collar.
    - Patients who are agitated because of another cause (for example traumatic brain injury) and will not remain still on the stretcher – it is appropriate to remove the c-collar (if required) and allow the patient to remain in a position which reduces any movement.
8. **Pediatrics**
    - Extrication of pediatrics should be made using a Pedi-board, LBB or KED as appropriate for the patient size.
    - Motion restriction techniques (such as c-collars and backboards) in pediatrics may cause a level of distress and increase patient movement.
    - If the patient becomes distressed and movement is increased, consider removing the c-collar, head-blocks and Pedi-Board to ensure the patient remains still.
    - If a pediatric patient is in a car seat, and no life-threatening interventions are required, keep the patient in the car seat (providing it is undamaged) and restrict movement in the car seat using blankets/towels for padding.
    - If the c-collar does not fit the patient correctly, (e.g. if the patient is too small) it should be removed.
9. **If a C-Collar Cannot be Placed**
    - Provide inline stabilization and support the head using the head block plate and head blocks – Forehead and Chin straps are not required.
    - Secure the patients torso using stretcher seat belts to avoid a fulcrum/pivot effect on the neck.

Patient Safety Considerations

- **Elderly Patients** (>65 years):
    - Increased risk of spinal fractures due to osteoporosis; apply SMR liberally if mechanism suggests injury.
    - Use extra padding to prevent pressure sores and ensure comfort during immobilization.
- **Pediatric Patients** (<13 years):
    - Use pediatric-sized cervical collars and boards to ensure proper fit.
    - Involve a parent or chaperone during assessment to reduce anxiety and provide history.
    - Apply SMR cautiously, as pediatric spines are more flexible but still vulnerable.
- **SMR Complications**
    - Avoid prolonged spinal board or scoop stretcher use to prevent pressure ulcers. If the patient is not time-critical, consider removal of the scoop stretcher once the patient is on the ambulance stretcher.
    - Ensure cervical collar does not restrict airway or jugular venous return.
- **Neurogenic Shock:**
    - Monitor for hypotension and bradycardia; avoid over-resuscitation with fluids to prevent pulmonary edema.

Clinical Insights & Additional Guidance

- **Extrication Devices**
    - Back boards (LBB) should be used as extrication devices only – patients should not be routinely transported to hospital on a back board.
    - Scoop stretchers are preferred to facilitate patient movement – where possible, the scoop stretcher should be removed once the patient is on the ambulance stretcher.
    - It is permitted to leave the LBB or scoop stretcher in place if immediate interventions (such as airway management) are required.
- **Handover Communication**
    - If SMR is indicated, and a c-collar has not been placed, clearly communicate the requirement for SMR with receiving hospital staff as part of IMIST-AMBO handover.

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: Description of trauma mechanism or reason for suspected spinal injury.
- Injuries / Information: Findings from spinal palpation, neurological assessment, and application of SMR.
- Signs: Vital signs, AVPU/GCS, presence of neurological deficits, SMR clearance criteria findings.
- Treatment: Type of spinal immobilization used (c-collar, board, scoop), method of extrication, airway management if applicable.
- Allergies: Relevant if medications were administered (e.g., for pain or sedation).
- Medications: Relevant if analgesia or sedation used during SMR application or extrication.
- Background: Medical history affecting SMR decision (e.g., osteoporosis, prior spinal injury).
- Other: Communication with receiving facility, documentation of reasons for deviation from standard SMR if applicable.

Key Performance Indicators

1. Assess and document spinal motion restriction assessment requirements.

References

1. Academic Emergency Medicine (1998) 'Out-of-hospital spinal immobilization effect', 5(3), pp. 214–219. doi:10.1111/j.1553-2712.1998.tb02615.x.
2. Prehospital Emergency Care (2019) 'No increase in SCI after switching to SMR'
3. Journal of Trauma and Acute Care Surgery (2014) 'Pressure ulcers from spinal immobilization: a systematic review', 76(4), pp. 1131–1141. doi:10.1097/TA.0000000000000153.
4. Annals of Emergency Medicine (1991) 'Respiratory effects of spinal immobilization in children', 20(9), pp. 1017–1019. doi:10.1016/S0196-0644(05)82983-2.
5. Prehospital Emergency Care (2012) 'Adverse effects of spinal immobilization in children', 16(4), pp. 513–518. doi:10.3109/10903127.2012.689925.
6. Journal of Trauma and Acute Care Surgery (2010) 'Spine immobilization in penetrating trauma: more harm than good?', 68(1), pp. 115–120. doi:10.1097/TA.0b013e3181c9ee58.
7. Journal of Trauma and Acute Care Surgery (2018) 'Prehospital spine immobilization/spinal motion restriction in penetrating trauma', 84(5), pp. 736–744. doi:10.1097/TA.0000000000001764.
8. Prehospital Emergency Care (2018) 'Spinal Motion Restriction in the Trauma Patient - A Joint Position Statement', 22(6), pp. 659–661. doi:10.1080/10903127.2018.1481476.
9. Journal of Emergency Medicine (2011) 'Cervical collars are insufficient for unstable cervical spine injury', 41(5), pp. 513–519. doi:10.1016/j.jemermed.2011.02.001.
10. Annals of Emergency Medicine (1984) 'Cervical spine injury model evaluation', 13(8), pp. 584–587. doi:10.1016/S0196-0644(84)80522-4.
11. Journal of Trauma Nursing (2006) 'Incidence of skin breakdown with cervical collars', 13(4), pp. 198–200. doi:10.1097/00043860-200610000-00007.
12. Injury (1996) 'Effect of rigid collar on intracranial pressure', 27(9), pp. 647–649. doi:10.1016/S0020-1383(96)00102-7.
13. British Journal of Anesthesia (2005) 'Mouth opening reduction with cervical collars', 95(3), pp. 344–348. doi:10.1093/bja/aei190.
14. Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine (2019) 'The Norwegian guidelines for prehospital management of adult trauma patients', 27(1), p. 2. doi:10.1186/s13049-018-0575-x.
15. ANZCOR (2015) Guideline 9.1.6 - Management of Suspected Spinal Injury. Available at: www.resus.org.au/guidelines/anzcor-guideline-9-1-6 (Accessed: 13 May 2025).
16. Annals of Emergency Medicine (2009) 'Out-of-hospital validation of the Canadian C-Spine Rule', 54(5), pp. 663–671.e1. doi:10.1016/j.annemergmed.2009.03.008.
17. Prehospital Emergency Care (2006) 'Comparison of the Ferno Scoop Stretcher with long backboard', 10(1), pp. 46–51. doi:10.1080/10903120500373373.

18. Emergency Medicine Journal (2003) 'Comparison of spinal board and vacuum mattress', 20(5), pp. 476–478. doi:10.1136/emj.20.5.476.